BTS (C) – V – 09 – 052 – I

B. Tech Degree V Semester Examination, November 2009

ME 502 METROLOGY AND MACHINE TOOLS

(2006 Scheme)

Time : 3 Hours | Maximum Marks : 100

PART - A

(Answer ***ALL*** questions)

(8 x 5 = 40)

I. (a) Distinguish between 'Line standard' and 'End standard'. Give examples.
(b) Explain the concepts of precision and accuracy.
(c) Describe gauges makers tolerance and wear allowance.
(d) Describe briefly the manufacture of slip gauges.
(e) Describe gauges and classification of gauges.
(f) With the help of a block diagram, explain how an engine lathe is specified.
(g) Explain boring and reaming.
(h) Explain the salient features of an NC machine.

PART – B

(4 x 15 = 60)

II. (a) Explain various sources of errors in measurement system. (10)
(b) A 200 mm sine bar is to be set to an angle of $28^\circ\ 5'\ 6''$. Find the height of the gauge blocks required using an M 87 set of gauge blocks. (5)

OR

III. (a) Why limits and fits used in manufacturing industry? Explain them. (10)
(b) Write a short note on tolerance and allowance. (5)

IV. (a) Describe the working of an interferometer with the help of a neat sketch. (10)
(b) Describe the working of clinometer and explain its uses. (5)

OR

V. (a) Describe various methods of assessing the surface texture. (10)
(b) How surface roughness is represented as per BIS? (5)

VI. (a) Explain various work holding devises used in machine tools. (10)
(b) Differentiate between Capstan and Turret lathe. (5)

OR

VII. (a) Explain hydraulic quick return mechanism used in shaping machine. (10)
(b) Calculate the minimum time to machine 250 mm long plate, 100 mm wide. Tool clearance at each end of stroke is 30 mm and 20 mm respectively. Cutting speed is 10 m/minute and return speed is 20 m/min. Feed is 1 mm per cycle and the distance the tool moves on either side of the work in width direction is 10 mm each. (5)

VIII. (a) Explain any one type of indexing used in milling machines. (10)
(b) Explain the working of optical dividing head used in milling operations. (5)

OR

IX. (a) Describe cylindrical and centre less grinding. (10)
(b) Explain the nomenclature of a Swiss type automatic screw machine. (5)

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